

A comparison between 125 Iodine brachytherapy and stereotactic radiotherapy in the management of juxtapapillary choroidal melanoma

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ABSTRACT

Aims To compare the treatment efficacy and radiation complications between 125 Iodine brachytherapy and stereotactic radiotherapy in the management of juxtapapillary choroidal melanoma.

Methods Consecutive juxtapapillary melanoma patients treated with radiotherapy were included. Patients were divided into two cohorts: patients treated with 125 Iodine brachytherapy and patients with stereotactic radiotherapy. Comparison included the rates postradiotherapy local recurrence, secondary enucleation, metastasis and radiotherapy complications. Kaplan–Meier estimates were used to determine the actuarial rates, and logrank test to compare between the estimates.

Results We included 94 patients with juxtapapillary melanoma treated with radiotherapy. The brachytherapy cohort included 30 patients and stereotactic radiotherapy was 64. The median follow-up was 46 months in both cohorts. No statistically significant differences existed between the two cohorts on comparing pretreatment clinical data and tumour characteristics. On comparing treatment efficacy, the actuarial rates at 50 months for tumour recurrence were 11% and 7% ($p=0.61$), secondary enucleation was 11% and 21% ($p=0.30$) and for metastasis were 4% and 16% ($p=0.11$), respectively. On comparing treatment complications, the actuarial rates at 50 months for cataracts were 62% and 75% ($p=0.1$), for neovascular glaucoma 8% and 47% ($p=0.002$), for radiation retinopathy 59% and 89% ($p=0.0001$), and for radiation papillopathy 39% and 74% ($p=0.003$), respectively.

Conclusions Both 125 Iodine brachytherapy and stereotactic radiotherapy demonstrate comparable efficacy in the management of juxtapapillary choroidal melanoma. However, stereotactic radiotherapy shows statistically significant higher radiation-induced ocular morbidities at 4 years postradiotherapy.

INTRODUCTION

The optic nerve, being anatomically wider in its extraocular diameter as compared with the optic disc, can be a physical obstacle for complete coverage of the entire base of a juxtapapillary melanoma by a round plaque. The optic nerve may physically cause sliding of the posterior edge of a round plaque along the nerve with subsequent tilting away of the plaque from the sclera during plaque placement, resulting in tumour under treatment. Additionally, during plaque insertion in a juxtapapillary location, there is a potential risk for mechanical compromise of the optic nerve and its vasculature that may result in traumatic

visual loss. Various radiotherapeutic techniques for management of juxtapapillary choroidal melanomas have been employed including application of notched and slotted radioactive plaques, charged particle radiotherapy, and external beam stereotactic radiotherapy.^{1–8} In our institution, stereotactic radiotherapy has been frequently used as a non-invasive treatment for patients of juxtapapillary melanoma. However, some of these patients who declined or were not physically suitable for stereotactic radiotherapy were treated with iodine brachytherapy, provided that their tumours demonstrated adequate radiation dose coverage on pretreatment brachytherapy planning. In this report, we compare the efficacy and side effects of radiotherapy between two highly comparable cohorts of patients of juxtapapillary melanoma: one cohort was treated with iodine brachytherapy and the other with stereotactic radiotherapy.

METHODS

A retrospective electronic charts review was conducted for all consecutive patients with juxtapapillary choroidal melanoma who were treated with radiotherapy, between October 1998 and January 2006, at the Ocular Oncology department and Radiation Oncology departments of Princess Margaret Hospital/University Health Network (UHN), Toronto, Ontario, Canada. The UHN research ethics board approval was obtained for this review. Patients were divided into two groups: patients treated with brachytherapy and those treated with stereotactic radiotherapy. No other treatment modality, for example, transpupillary thermotherapy, has been used for any of the included patients.

Juxtapapillary location

The optic nerve is 1.5–2 mm larger in diameter on the episcleral side from the optic disc. Following the Collaborative Ocular Melanoma Study (COMS) recommendations,⁹ the episcleral plaques used were designed to be at least 4 mm larger than the largest tumour diameter to create a radiation field that allows for errors in plaque placement, plaque movement and uncertainties in the location of tumour edge. Thus, all round episcleral plaques inserted to treat choroidal melanoma located at 2.5 mm or less from the optic disc should be virtually touching the episcleral portion of the optic nerve. Subsequently, a melanoma was considered to be juxtapapillary if the posterior tumour margin was 2.5 mm or less from the optic disc, touching the optic disc, or partially or completely surrounding the disc.

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Patient data included demographics, medical and ocular history, pretreatment findings based on clinical ophthalmological examination, fundus photography, ultrasonography, and fluorescein angiography. Follow-up examinations data included post-treatment tumour measurements, screening for metastases and treatment-related complications.

Treatment efficacy measures for both cohorts included the rates of postirradiation local tumour recurrence, secondary enucleation (for management of tumour recurrence or intractable radiotherapy-related complications) and melanoma-related systemic metastasis. Guided by the COMS, tumour recurrence was defined as total regrowth of at least 30% of tumour thickness or at least 500 μm of tumour base that was recorded within two successive post-treatment visits to exclude cases of transient postirradiation tumour swelling secondary to radiation vasculopathy-induced haemorrhage.

Radiation-induced complications included radiation cataract, neovascular glaucoma (NVG), radiation retinopathy and radiation optic neuropathy.

Radiation cataract was defined as the development of significant lens opacity in the irradiated eye as compared with the

opposite eye. NVG was diagnosed if there was neovascularisation of the iris or angle associated with elevated intraocular pressure. Radiation retinopathy was defined as the presence of clinical or angiographic evidence of spreading retinal ischaemic vasculopathy, more than 2 mm away from the treated tumour margin, with or without macular involvement, manifesting as retinal haemorrhage, exudates, infarcts, oedema or neovascularisation. Presence of at least two of the previous findings was required to record the development of radiation retinopathy. Radiation papillopathy was defined as the development of optic disc swelling with peripapillary haemorrhages and exudates, or disc pallor.

¹²⁵Iodine brachytherapy

Melanomas managed with brachytherapy in this study were treated using the standard (COMS) round, gold-shielded, episcleral plaques. The plaque diameter was at least 4 mm larger than the tumour's longest basal dimension. Guided by dosimetry plaque simulator software (BEBIG, Germany), the ¹²⁵Iodine seeds were either evenly loaded with single activity seeds, and/or asymmetrically loaded with different activities seeds ranging from 0.5 to 7 millicurie (figure 1). Asymmetric loading, in this study,

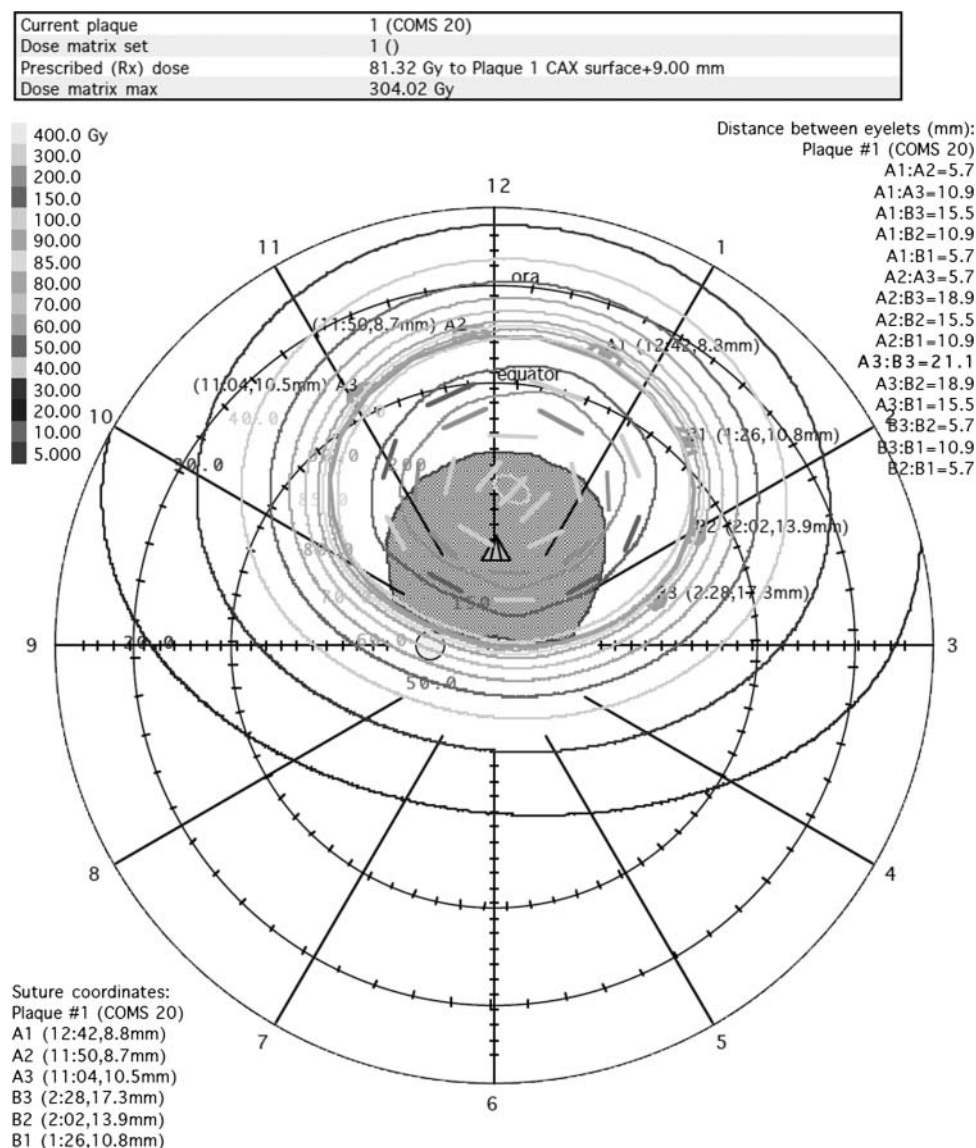


Figure 1 Pretreatment ¹²⁵Iodine brachytherapy planning for a juxtapapillary choroidal melanoma with an asymmetrically loaded plaque illustrating the isodose curves with sufficient radiation dose coverage of the juxtapapillary tumour margin.

entails inserting higher energy seeds at slots nearer to the posterior plaque edge. This would push the radiation penumbra more posteriorly, and can include a thin posterior tumour margin adjacent to the optic disc within the radiation field. Due to plaque placement limitation, almost 1 mm margin coverage could be achieved at the nerve side, beyond plaque edge, with dose coverage on average ~15% lower than tumour apex. Based on COMS protocol, all melanomas in this study were treated to a total dose of 8500 centigray (cGy) delivered to the tumour apex at a mean dose rate of 50 cGy/h (range=49.5–52.2 cGy/h). Plaques in this study were inserted under general anaesthesia. To facilitate plaque insertion, recti muscles may be disinserted at tumour base. An extra 1 mm is usually added to tumour height, on calculating apex dose, to compensate for possible intrusion of a part of an oblique muscle between the plaque and sclera. Proper plaque placement was verified intraoperatively using an illuminated sclera indenter after plaque fixation.

Stereotactic radiotherapy

Our treatment technique has been previously described.¹ The prescribed total radiation dose was 70 Gy delivered in five fractions with treatment every alternate day over 10 days of 6 MV photons. All patients were CT-simulated supine and immobilised using the Radionics Gill-Thomas-Cosman stereotactic relocatable head frame (Integra Radionics, Burlington, Massachusetts,

USA) adapted with an inhouse adjustable eye-monitoring device. The radiation therapists monitor eye position via the eye-monitoring device video camera, and the inhouse developed software registers the eye position and detects deviations. If deviation occurs, treatment is stopped, the patient is allowed to rest and treatment then restarts. Treatment planning was accomplished using the Radionics XKnife-4, and more recently XKnife RT3 software (Integra Radionics) (figure 2). Multiple converging non-coplanar arcs of radiation were generated using circular collimators ranging from 10 to 27.5 mm. Radiotherapy was delivered using Radionics circular collimators and the Linac couch-mount assembly system mounted on a Varian 2100C (Varian Inc., Palo Alto, California, USA) Linear accelerator, and more recently the Elekta Synergy S unit (Elekta, Crawley, UK).

Statistical analysis

Patient overall survival was calculated based on Kaplan–Meier estimates. Rates of complications were calculated using the cumulative incidence approach. Risk factors for metastasis were tested using the Cox proportional hazard model. When the variables were continuous, they were dichotomised at the median. Logrank test was used to compare the statistical significance between the survival curves in both cohorts. Results were considered significant if the p value was less than or equal to 0.05.

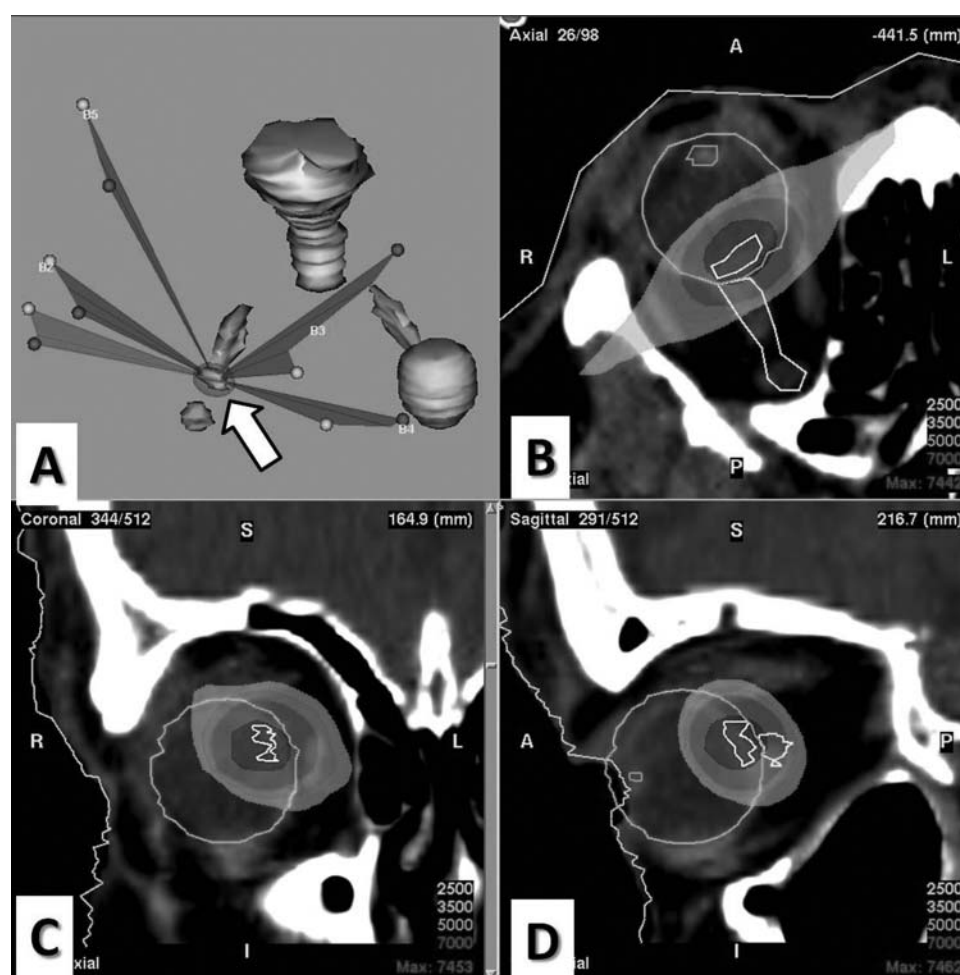


Figure 2 (A) Three-dimensional view of the stereotactic radiotherapy treatment plan illustrating the relationships of the five non-coplanar beams focusing at the juxtapapillary melanoma (arrow) with the nearby ocular and neurological structures; (B) axial; (C) coronal; (D) sagittal CT images through the juxtapapillary tumour illustrating the isodose distribution with complete radiation coverage of the tumour with 70 Gy.

The statistical analyses were performed using V9.1 of the SAS system (SAS Institute, Cary, North Carolina, USA).

RESULTS

Pretreatment patient data

The brachytherapy cohort included 30 patients with the posterior tumour margins located 0–2.5 mm from the optic disc. The stereotactic radiotherapy cohort included 64 patients with posterior tumour margins located 0–2.1 mm. The two cohorts were found to be highly comparable as there were no statistically significant differences between them in terms of demographics, pretreatment medical conditions that may confound the post-treatment outcome and tumour dimensions. Except for two patients treated with stereotactic radiotherapy, other patients presented with circumpapillary melanoma during the period of this study were treated by primary enucleation (table 1).

Treatment outcome

Post-treatment median follow-up for the brachytherapy group was 46 months (mean 51 months, range 12–136 months), and for stereotactic radiotherapy group was 46 months (mean 50, range 12–118). There was a minimal difference in the treatment outcomes between the two groups, which were not found to be statistically significant in this study. A larger sample size may be needed to further validate these results. Three patients (10%) of the brachytherapy cohort underwent enucleation for local tumour recurrence. Six patients (10%) of the stereotactic radiotherapy cohort underwent enucleation, in two patients (3%) for local recurrence and in 4 (6%) for intractable NVG (table 2).

Complications

No intraoperative or immediate postoperative complications occurred in any patient. Actuarial rates of all late radiation-related complications were calculated at 50 months post-treatment in both cohorts. There were no statistically significant differences between the two cohorts in the incidence rates of radiation cataract. However, statistically significant greater incidence rates, post stereotactic radiotherapy, were observed for

Table 1 Comparison of pretreatment patient data, tumour characteristics and the relation of posterior tumour margins to the disc between the two cohorts

	Brachytherapy (n=30)	Stereotactic radiotherapy (n=64)	p Value
Age	59 (58, 39–77)*	63 (62, 28–89)	0.11
Sex (M : F)	15 (50%):15 (50%)	36 (56%) : 28 (44%)	0.57
Diabetes	3 (10%)	3 (5%)	0.33
Hypertension	7 (23%)	18 (28%)	0.62
Tumour height	4.3 (4.5, 1.5–9.1)	4.2 (4.6, 1.5–11)	0.98
Tumour base	10.0 (11.0, 6–16)	9.9 (10.2, 4.7–17)	0.18
Tumour edge to disc distance	0–2.5 mm	0–2.1 mm	
Tumour edge to disc relationship			
Not touching the disc	15 (50%)	22 (34%)	
>0° to <180°	14 (47%)	36 (57%)	
>180° to <360°	1 (3%)	4 (6%)	
360°	0	2 (3%)	
Follow-up (months)	46 (51, 6–105)	46 (50, 6–118)	0.46

*Median (mean, range).

Table 2 Comparison of the actuarial rates of the treatment outcomes at 50 months post radiotherapy

	¹²⁵ Iodine brachytherapy (n=30)	Stereotactic radiotherapy (n=64)	HR	p Value
Tumour recurrence	11% (3%–39%)*	7% (3%–18%)	1.47	0.61
Secondary enucleation	11% (3%–40%)	21% (12%–35%)	0.51	0.30
Systemic metastasis	4% (1%–25%)	16% (9%–29%)	0.21	0.11

*(95% confidence interval).

NVG, radiation retinopathy and papillopathy in comparison with brachytherapy (table 3).

DISCUSSION

The uncertainty regarding the relationship of the plaque to the optic nerve and tumour base has been illustrated in a study on cadaver eyes where the posterior edges of plaques applied in juxtapapillary locations were visualised by MRI to be tilted away from the optic nerve.¹⁰ This plaque tilt was found to cause reduction greater than 10% in the prescribed radiation dose to the tumour apex in 23% of patients, which potentially may compromise tumour control.^{11 12} In another report based on intraoperative echographic localisation of plaque placement, there was incomplete coverage of at least one margin of the posterior uveal melanomas in 21% of cases treated by episcleral plaques, when using the standard localisation techniques.¹³ Although notched episcleral plaques are designed to better accommodate the optic nerve, they still carry a potential risk for damage to both the optic nerve and its vasculature during insertion similar to conventional round plaques.¹⁴ Finger's slotted plaque, however, having no rim for the slot, with 8 mm slot width, and variable slot depth, may effectively encircle the optic nerve with its sheaths and provide a radiation field surrounding the optic nerve that can completely encompass a circumpapillary melanoma without geographic miss.¹⁵

In our institution, stereotactic radiotherapy, as a non-invasive radiotherapy procedure, is elected to treat juxtapapillary melanoma to minimise those potential physical risks associated with juxtapapillary plaque insertion. We have developed an inhouse technique for accurate localisation, and we adopted a treatment dose and fractionation scheme akin to those used in some proton beam radiotherapy protocols for choroidal melanoma.^{6 16 17}

Table 3 Comparison of the actuarial rates of treatment complications between the two cohorts

	IBT (n=30)	SRT (n=64)	HR	p Value
Radiation cataract	62% (42%–83%)*	75% (59%–88%)	0.63	0.1
Neovascular glaucoma	8% (2%–29%)	47% (35%–62%)	0.20	0.002
Radiation retinopathy	60% (41%–79%)	89% (78%–96%)	0.29	0.0001
Radiation papillopathy	40% (24%–63%)	74% (61%–86%)	0.37	0.003

*(95% confidence interval). IBT, ¹²⁵Iodine brachytherapy; SRT, stereotactic radiotherapy.

Brachytherapy and stereotactic radiotherapy are fundamentally different such that brachytherapy inherently results in more conformal dose distributions, and delivers the radiation continuously at a low radiation dose rate. This continuous low dose rate exposure is radiobiologically advantageous in minimising the potential for late adverse effects on the surrounding normal tissues.¹⁸ Stereotactic radiotherapy is based on higher dose rate and high dose per fraction radiation that is radiobiologically designed to maximise tumour control. However, the doses delivered are toxic to the surrounding normal tissues. It is the steep dose gradients inherent to stereotactic technology that minimises dose exposure to the surrounding normal tissues aiming to maintain acceptable toxicity rates. Considering tumour localisation within stereotactic irradiation field, ocular melanomas are considered a small target volume for dose distribution conformity similar to brachytherapy to be achieved, and the surrounding normal tissue would be exposed to greater dose. This is the reason why, as a means to reduce the potential for late effects with stereotactic radiotherapy, our practice has been to treat on alternate days to maximise the time for tissue repair of the surrounding structures in between fractions.¹⁸

Comparison of the treatment efficacy between brachytherapy and stereotactic radiotherapy for juxtapapillary melanoma in the two treated groups showed highly comparable outcomes despite their small sample size. Nevertheless, these rates were also comparable with those published in a study of management of 650 juxtapapillary melanoma patients with brachytherapy using various radioactive isotopes employing round and notched plaques, where the estimated rates of tumour recurrence and metastasis at 5 years were 14% and 11%, respectively.¹⁹ On the other hand, our tumour recurrence rate with stereotactic radiotherapy was 7% at 4 years, which may be relatively lower than the reported rate of 24% at 5 years using stereotactic irradiation for juxtapapillary melanoma with dose of 60 Gy in 10 fractions.²⁰

With regard to radiotherapy complications, there was no statistically significant difference between the two groups in terms of radiation cataract. In theory, anterior segment structures should receive higher radiation dose with stereotactic radiotherapy, which is delivered through the front of the eye in comparison with brachytherapy. However, the equivalent high risk of cataract development with both modalities as observed in this study likely reflects the lower threshold of lens sensitivity to radiation exposure.^{21 22}

Irradiation of tumours close to the optic nerve is generally associated with substantial visual loss, irrespective of the type of radiotherapy.^{1 3–6 19 20 23} This is attributed to higher dose reaching the optic nerve head and its vascular supply, with subsequent development of radiation papillopathy, or from direct radiation damage to the nearby fovea or papillomacular bundle. Moreover, proximity of posterior tumour margin to the optic nerve or the macula has been reported as a risk factor for development of delayed onset radiation retinopathy with subsequent significant visual loss.^{24 25}

This study has demonstrated that with iodine brachytherapy and stereotactic radiotherapy there were high rates of radiation retinopathy and papillopathy, in comparison with reported rates post radiotherapy of melanoma in other locations within the ocular fundus.^{26–32} Nonetheless, radiation retinopathy and papillopathy rates were significantly higher with stereotactic radiotherapy when compared with brachytherapy. Our preliminary observations suggest that inclusion of a wider area of normal tissues within the planned radiation field of stereotactic radiotherapy, in addition to the difference in the radiobiological

effects resulting from the difference in the dose rates between these two techniques, may explain the higher incidence of radiation-induced morbidities with stereotactic irradiation. The relatively lower complication rates in the brachytherapy cohort presumably reflects the greater conformity of the dose distribution, the sharper dose decline beyond the plaques edge with lower dose exposure to the adjacent critical structures and the effect of a lower dose rate as compared with stereotactic radiotherapy. Nonetheless, a complete dosimetric analysis for comparison of the intraocular radiation dose distribution and its relationship to the rates of complications in these cases is currently in progress.

In this study, stereotactic radiotherapy has demonstrated statistically significant higher rate of NVG as compared with brachytherapy. This high rate of NVG was previously reported with charged particle radiotherapy.^{33 34} NVG may be related to the greater ischaemic changes occurring at the posterior segment, rather than direct radiation insult to the anterior segment vasculature associated with stereotactic radiotherapy. It is postulated that extensive radiation-induced ischaemic retinopathy produces proangiogenic factors, such as vascular endothelial growth factor, which diffuses through the vitreous to induce vascular proliferative changes in the iris and angle and subsequent development of NVG.³⁵

In summary, despite the retrospective and non-randomised structure of this study, it reveals that when comparing these two radiotherapy methods of photon irradiation for treatment of juxtapapillary melanoma, both ¹²⁵Iodine brachytherapy and stereotactic radiotherapy have shown comparable rates of local recurrence, secondary enucleation and metastasis. Nevertheless, the posterior segment radiation-related morbidities were significantly higher with stereotactic radiotherapy in its current dose and fractionation scheme when compared with brachytherapy at 4 years after radiotherapy.

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Competing interests None.

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